

Human-in-the-Loop Egyptology: A System for Ancient Egyptian Text Study

Innokentiy Humonen
AXXX

Moscow, Russian Federation
g1410ish@gmail.com

Maksim Golyadkin
HSE University

Moscow, Russian Federation
AXXX
Moscow, Russian Federation
mgolyadkin@hse.ru

Valeria Rubanova
ITMO University

Saint Petersburg, Russian Federation
467271@niuitmo.ru

Ekaterina Alexandrova
HSE University
Moscow, Russian Federation
ev.aleksandrova@hse.ru

Alexandre A Loktionov
HSE University
Moscow, Russian Federation
University of Cambridge
Cambridge, United Kingdom
aloktionov@hse.ru

Ilya Makarov
AXXX
Moscow, Russian Federation
Trusted AI Center
RAS
Moscow, Russian Federation
iamakarov@hse.ru

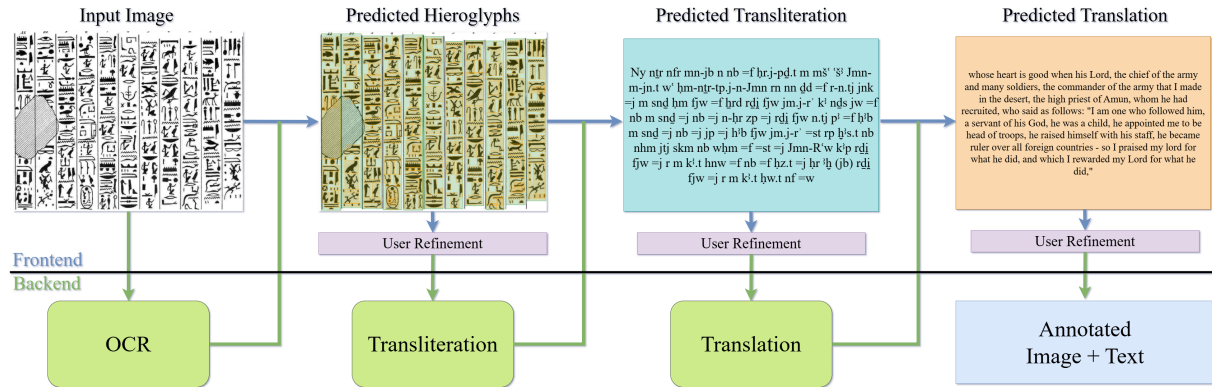


Figure 1: System workflow showing the transformation from input image to final annotation. The backend pipeline handles recognition, transliteration, and translation, while the frontend enables user refinement to accelerate the annotation process.

Abstract

We present a prototype web-based human-in-the-loop system for studying Ancient Egyptian hieroglyphic texts that integrates an image-to-text pipeline into an interactive workspace for iterative refinement. In a formative user study with Egyptologists and students, the system enabled faster work and higher-quality results than a manual workflow.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Computer vision tasks**; Natural language processing.

Keywords

Egyptology, hieroglyphs, human-in-the-loop, interactive systems, OCR, transliteration, translation

ACM Reference Format:

Innokentiy Humonen, Maksim Golyadkin, Valeria Rubanova, Ekaterina Alexandrova, Alexandre A Loktionov, and Ilya Makarov. 2026. Human-in-the-Loop Egyptology: A System for Ancient Egyptian Text Study. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798955>

1 Introduction

Ancient Egyptian hieroglyphic texts preserve rich evidence about language, religion, administration, and everyday life. However, working with these texts remains slow and difficult: many sources are available primarily as images, and limited digitization makes search, lookup, and reuse far less accessible than for modern languages.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798955>

Recent deep learning work suggests strong potential to assist Egyptological analysis. NLP-oriented models have been proposed for tasks such as transliteration and translation in the text domain [1], yet practical workflows often start from images rather than clean digital text. More recently, an end-to-end pipeline mapping images to textual domain has been introduced [3], but without a user interface it remains difficult to adopt for most Egyptologists.

In this work, we bridge this gap by presenting a prototype web-based system that brings an image-to-text pipeline into a web-based workspace for Egyptological practice. The system produces codes, transliteration, and translation as explicit intermediate outputs that users can inspect, correct, and iteratively refine, making model assistance accessible without specialized ML expertise. We evaluated the prototype in a formative user study and found that it supports faster work with improved output quality compared to a manual workflow.

Our contributions are threefold: the first interactive Egyptological analysis system to embed an image-to-text deep learning pipeline, interface mechanisms for iterative refinement through explicit intermediate representations and on-demand pipeline stages, and a formative study with demonstrating improved efficiency and quality.

2 Background and Related Work

Working with a hieroglyphic inscription involves a multi-step workflow in which Egyptologists identify signs, transliterate the text, and then translate it into a modern language. Sign lookup is often the most difficult part of this process: the space of possible hieroglyphs is large, and memorizing the full inventory is unrealistic. Standardized encodings such as Gardiner codes (e.g., D36, G1) provide a shared way to refer to signs [2], but scholars still need to consult reference materials extensively. This need does not end with sign identification: transliteration and translation also require repeated checking of dictionaries and related resources to resolve uncertain readings. While digital references exist, they remain limited in coverage, and scholars therefore continue to rely on printed materials. As a result, the workflow is slow and fragmented, motivating both further digitization and assistive tools for Egyptological text processing.

Existing interfaces offer only partial support. General-purpose annotation tools (e.g., Label Studio [9]) are convenient for image-based OCR annotation, but they provide no native support for Egyptian hieroglyphs such as sign inventories, Gardiner-code entry, or faithful glyph rendering. Conversely, domain-specific tools such as JSesh [8] support hieroglyphic text entry and rendering, but are primarily standalone desktop applications and are not designed for annotation workflows. In practice, this forces scholars to switch between tools, which is slow and hinders large-scale digitization.

Recent machine-learning work has proposed models for hieroglyphic text analysis [1, 7], but many focus on text-only settings and therefore depend on digitized inputs, which remain limited. While image-to-text pipelines exist [3], they are typically research prototypes without an interactive UI for markup and correction. As a result, Egyptologists still lack a widely available out-of-the-box tool for image-based hieroglyphic text analysis in daily work.

We present a prototype that unifies hieroglyphic text analysis in a single workspace, integrating sign encoding and visualization with AI assistance for interpretation. By reducing tool switching and supporting iterative refinement, the system is designed for everyday scholarly use and to demonstrate to the Egyptology community that modern deep learning can meaningfully support their research.

3 Human-in-the-Loop Egyptology System

We propose a prototype system for human-in-the-loop Egyptology that combines an interactive workspace with DL-based suggestions for multistep text analysis. We first describe the design goals that guided the prototype and then explain how these goals are implemented through the system components and workflow.

3.1 Design Goals

We translate the workflow constraints described in the previous section into four requirements that guide system design. Rather than targeting full automation, we aim to support Egyptologists' manual workflow by mimicking familiar steps with DL assistance to reduce routine effort and improve accuracy, while keeping expert judgment central.

(i) **Reduce context switching.** The system should co-locate the key resources needed during analysis (image context, sign encodings, lexicon access) in a single workspace. This goal prioritizes minimizing copy-paste and manual re-entry, and maintaining tight alignment between visual evidence and linguistic representations.

(ii) **Accelerate routine work with DL assistance.** The system should incorporate DL pipelines as assistive suggestions that speed up routine steps. Users must be able to refine suggestions easily.

(iii) **Support iterative verification via explicit intermediate representations.** Because suggestions can be wrong and interpretations evolve, the system should make intermediate representations explicit (Gardiner codes, transliteration, translation) and easy to inspect, compare, and edit.

(iv) **Capture data as a byproduct of use.** The system should export structured results and expert corrections in standard formats, enabling reuse for future model improvement (with explicit user consent).

Together, these goals motivate an integrated human-in-the-loop workflow that combines interactive tooling with DL-based suggestions. We next describe the implementation that puts these goals into practice.

3.2 System Overview

Our prototype builds on Label Studio as a text-study workspace for Ancient Egyptian inscriptions: users work from images and iteratively produce Gardiner codes, transliteration, and translation in one environment. Intermediate representations remain visible and editable, supporting verification of model predictions and expert interpretation. We connect the DL pipeline via Label Studio's *ML Backend*, separating a suggestion service from a web frontend where users review and correct the current analysis state (Figure 1).

Text Study Workspace. The workspace supports OCR-style region editing, where users place bounding boxes and fill structured fields (codes, transliteration, translation). In our use case, this

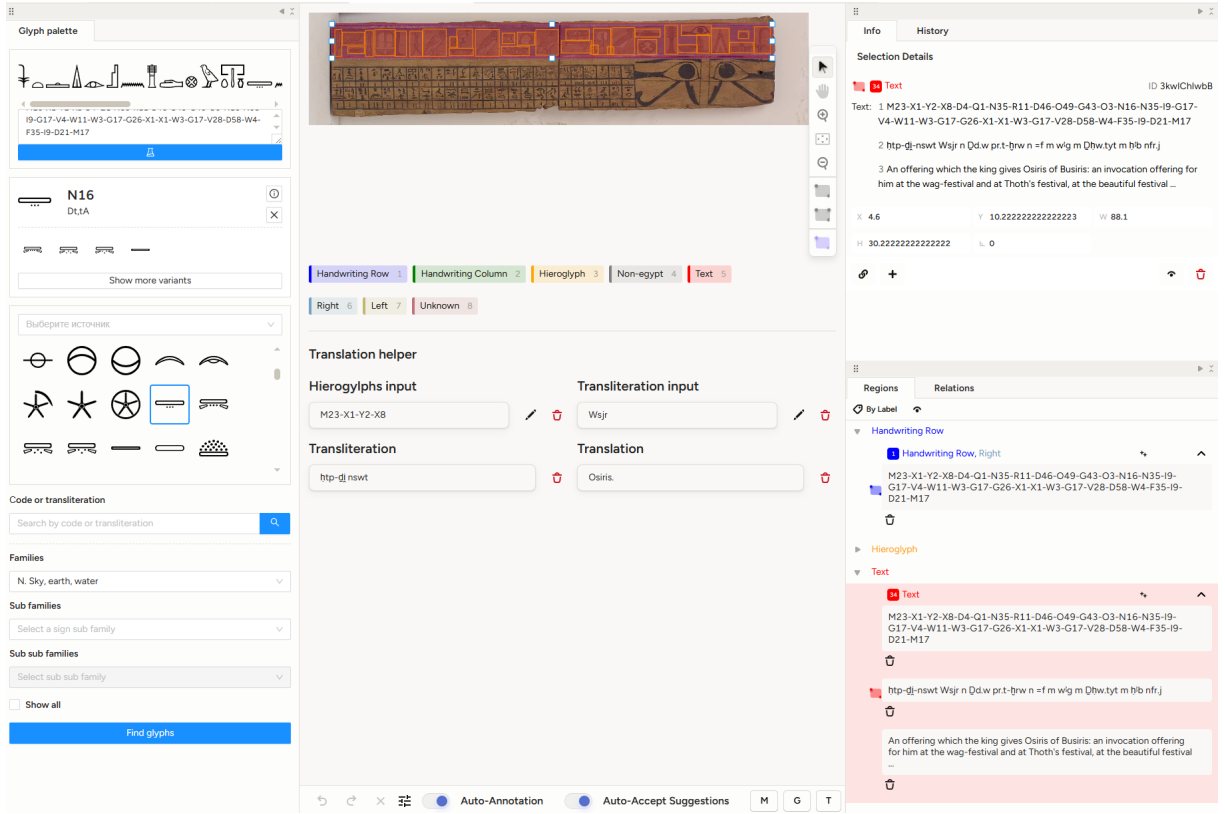


Figure 2: Interface of the proposed system for studying hieroglyphic texts. The center panel shows a text image; the left panel supports fast lookup and entry of Gardiner codes via a *Hieroglyph Palette* and *Renderer*; the right panels expose editable codes, transliteration, and translation for iterative refinement of model suggestions.

anchors linguistic hypotheses to image evidence and enables step-by-step refinement across representations. We extend the default Label Studio interface with two Egyptology-specific components (Figure 2): the *Hieroglyph Palette* for fast lookup and insertion of Gardiner codes, and the *Hieroglyph Renderer* for immediate visualization of code sequences inside the workflow. Finally, we add an interactive *Translation Helper* panel (Google Translate-style) that takes Gardiner codes and/or transliteration as input and returns suggested transliteration and English translation for quick reuse in the corresponding fields.

ML Backend. The ML Backend utilizes a modular DL pipeline that generates editable suggestions for multiple stages of text study. Given an input image, the pipeline first proposes sign-level outputs as Gardiner codes with associated regions and orders them into a reading-order sequence. It then produces candidate transliteration and translation using seq2seq Transformer models. We employ the multistage pipeline introduced in [3]. While the pipeline’s individual stages achieve competitive performance when evaluated separately (as reported in the referenced work), end-to-end predictions remain imperfect [4]. However, this matches the multistep nature of Egyptological practice, where working through intermediate representations support user-guided correction.

Technical Details. The system is designed to run on standard hardware. In the current implementation, generating initial suggestions requires approximately 8 GB of RAM and takes about 2 minutes per image on CPU or around 25 seconds on a consumer-grade GPU. The prototype is provided as a hosted web application, so it requires special hardware only from the hoster and can be accessed by the users from any standard browser.

3.3 Human-in-the-Loop Workflow

We extend the Label Studio interface with control buttons that invoke individual stages of our multi-step DL pipeline on demand, enabling an iterative loop of prediction, inspection, correction, and re-running downstream stages.

In a typical session, a user loads an image and presses the first button to run sign detection and recognition, producing line regions and a reading-order sequence of Gardiner codes linked to image regions. After correcting codes as needed, the second button generates a draft transliteration, and the third button generates a draft translation from the current intermediate representations. If later-stage outputs reveal an earlier mistake, users revise the upstream representation and re-run the relevant stage(s) to update predictions. For finer-grained interaction, the *Translation Helper* supports quick checks of individual words or short phrases. When finished,

Table 1: Output quality under different assistance levels. Values (reported as percentages) are mean $\pm$ std. The system achieved the best recognition and translation quality, suggesting interactive assistance can outperform fully automatic or human-only workflows. The scores are in line with the broader testing from [3] and [5].

Condition	Recognition (CER) $\downarrow$	Transliteration (chrF++) $\uparrow$	Translation (chrF++) $\uparrow$
Baseline (human-only)	10.6 $\pm$ 3.8	45.9 $\pm$ 6.5	41.3 $\pm$ 8.4
DL-only (end-to-end)	9.0 $\pm$ 1.3	61.0 $\pm$ 1.0	48.0 $\pm$ 1.1
DL + user-fixed inputs	9.0 $\pm$ 1.3	63.4 $\pm$ 3.8	50.8 $\pm$ 9.3
System (final)	7.2 $\pm$ 2.4	61.8 $\pm$ 6.9	51.8 $\pm$ 4.0

Table 2: Efficiency and editing effort. Task completion time (TCT, minutes) is reported for both conditions. Normalized Levenshtein distance (NLD, %, lower is less editing) is reported for the system condition. The system reduced total time with modest editing effort.

Subtask	TCT (min)		Edit Distance (NLD)
	Baseline	System	
Recognition	35	33	6.7
Transliteration	16	10	12.7
Translation	9	5	16.8
Total	60	48	–
Mean	20	16	12.1

users export results via Label Studio’s built-in functionality, preserving aligned regions, codes, transliteration, and translation for downstream reuse.

4 User Study

To understand how the prototype supports real Egyptological practice, we performed a formative user study comparing our system to a manual workflow. We measure both efficiency and correctness of the outputs, and quantify how users refine model suggestions. We also collect brief usability feedback to identify remaining bottlenecks in the system.

4.1 Study Setup

Participants. We recruited $N=8$ participants with Ancient Egyptian languages experience: 6 undergraduate Egyptology students and 2 PhD Egyptologists. All participants had prior training in the language. Participants were recruited via the authors’ professional and academic networks across multiple institutions and volunteered without compensation.¹ The sample size of $N=8$ participants, while modest by computational standards, reflects the practical constraints of working with a specialized and comparatively small community of Egyptologists.

Task. We used a single Middle Egyptian monument as the study material: the British Museum stela EA567 (available for non-commercial reuse under a CC BY-NC-SA 4.0 licence). To match common Egyptological practice, participants worked from a facsimile-style

drawing, and we prepared two cropped excerpts (approximately 2.5 lines each). The excerpts were chosen to be similar in difficulty and sufficiently independent such that working on one did not help with the other. For each excerpt, participants produced three outputs: a reading-order sequence of Gardiner codes, transliteration, and English translation. To score result quality, we used a generally accepted study as a reference [6].

Study design. Each participant completed two trials: they analyzed one excerpt with the baseline workflow and the other excerpt with our system. In the baseline condition, participants used their usual reference workflow (e.g., sign lists and dictionaries, often via PDFs/web resources and/or JSesh for rendering) without our prototype. In the system condition, they used our tool, with no restriction on additionally consulting external resources. The study was fully counterbalanced: both the starting condition (baseline-first vs. system-first) and which excerpt was paired with each condition were randomized, yielding four possible trial sequences. We assigned two participants to each sequence.

Procedure. Participants completed the study in their own setting using their usual setup. They first watched a 10-minute video demonstrating the system and submission process (no practice task). For each trial, participants submitted the three outputs via a Google Form and self-recorded time per output, followed by brief feedback.

Measures. We measured efficiency and output quality for both conditions. For efficiency, we report average task completion time (TCT) for each output and total TCT. For quality, we compare participant outputs to the reference: Gardiner code sequences are scored with sign-level character error rate (CER), and transliteration and translation are scored with chrF++. To quantify how participants refined model suggestions in the system condition, we compute Normalized Levenshtein Distance (NLD) between the model’s draft output and the participant’s submission.

Ethics. The study protocol was approved by the authors’ institutional ethics review process. All participants provided informed consent and could withdraw at any time. The data were anonymized for analysis.

4.2 Results

Quality metrics comparison. We report four levels of assistance: the baseline workflow (human-only), *DL-only* outputs from the pipeline, *DL with user-fixed inputs*, and the system outputs produced by participants with our tool. This breakdown separates raw model quality from the effects of human correction. Table 1 shows consistent quality gains over the baseline across all outputs. Notably, the final system results are comparable to participant-produced outputs,

¹We thank the AXXX development team led by S. Mamontov and all experiment participants.

suggesting that the pipeline can reach student-level performance on this material.

Time and editing effort comparison. Table 2 summarizes efficiency and the amount of editing required in the system condition. Overall, participants were faster with our system (48 vs. 60 minutes total). At the same time, users made relatively modest edits to model drafts, suggesting that the time savings were not offset by correction burden.

Qualitative feedback. Overall, participants were positively surprised that the model could recognize many hieroglyphs, including less common signs, and reported that this reduced manual lookup effort. They also described the transliteration and translation suggestions as generally useful, but noted that both often required small corrections. The main drawback raised by participants concerned the interface inherited from Label Studio: several found it not intuitive and encountered usability issues and occasional bugs, which reduced the overall experience.

Thus, this user study provides evidence that the prototype can support Egyptological text analysis: participants completed tasks faster with modest editing effort and produced higher-quality outputs than the baseline. At the same time, the results indicate that both the DL pipeline and the inherited interface need further improvement before the system can reliably support everyday use.

5 Conclusion and Future Work

We presented a prototype human-in-the-loop system for Ancient Egyptian text analysis. In a formative user study, participants completed tasks faster and produced higher-quality outputs than a baseline workflow, indicating that the approach can support expert practice and learning. We will continue development with feedback from Egyptologists, improving the DL pipeline with higher-quality models and redesigning the interface beyond Label Studio.

Acknowledgments

This work was supported by a grant, provided by the Ministry of Economic Development of the Russian Federation (agreement dated June 20, 2025 No. 139-15-2025-011, identifier 000000C313925P4G0002).

References

- [1] Mattia Cao, Nicola Cao, Angelo Colonna, and Alessandro Lenci. 2024. Deep Learning Meets Egyptology: a Hieroglyphic Transformer for Translating Ancient Egyptian. In *Proceedings of the 1st Workshop on Machine Learning for Ancient Languages (ML4AL 2024)*. Association for Computational Linguistics, Hybrid in Bangkok, Thailand and online, 71–86. doi:10.18653/v1/2024.ml4al-1.9
- [2] Morris Franken and Jan C. van Gemert. 2013. Automatic Egyptian hieroglyph recognition by retrieving images as texts. In *Proceedings of the 21st ACM International Conference on Multimedia (Barcelona, Spain) (MM '13)*. Association for Computing Machinery, New York, NY, USA, 765–768. doi:10.1145/2502081.2502199
- [3] Maksim Golyadkin, Innokentiy Humonen, Yanis Plevokas, Ekaterina Bureeva, Ekaterina Alexandrova, and Ilya Makarov. 2025. Automatic Interpretation of Ancient Egyptian Texts for Education and Research. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '25)*. Association for Computing Machinery, New York, NY, USA, Article 39, 3 pages. doi:10.1145/3721250.3743025
- [4] Maksim Golyadkin, Innokentiy Humonen, Valeria Rubanova, Danil Kalin, Ianis Plevokas, Dmitry Nikolotov, Aleksandr Utkov, Nikita Sidelnikov, Petr Ivanov, Ekaterina Bureeva, Ekaterina Alexandrova, and Ilya Makarov. 2025. MuMMY: Multimodal Dataset supporting VLM-based Egyptology Research Assistant. In *Proceedings of the 33rd ACM International Conference on Multimedia (Dublin, Ireland) (MM '25)*. Association for Computing Machinery, New York, NY, USA, 12875–12881. doi:10.1145/3746027.3758231
- [5] Innokentiy Humonen, Maksim Golyadkin, Danil Kalin, and Ilya Makarov. 2025. Automatic Image Translation of Long Ancient Egyptian Texts for Augmented Reality Applications. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE Institute of Electrical and Electronics Engineers, Daejeon, Korea, 875–876. doi:10.1109/ISMAR-Adjunct68609.2025.00234
- [6] Miriam Lichtheim. 1988. *Ancient Egyptian Autobiographies Chiefly of the Middle Kingdom: A Study and an Anthology*. Orbis Biblicus et Orientalis, Vol. 84. Universitätsverlag / Vandenhoeck & Ruprecht, Freiburg, Switzerland / Göttingen, Germany. doi:10.5167/uzh-141274
- [7] So Miyagawa. 2025. RAG-Enhanced Neural Machine Translation of Ancient Egyptian Text: A Case Study of THOTH AI. In *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, Mika Härmäläinen, Emily Öhman, Yuri Bizzoni, So Miyagawa, and Khalid Alnajjar (Eds.). Association for Computational Linguistics, Albuquerque, USA, 33–40. doi:10.18653/v1/2025.nlp4dh-1.4
- [8] Rosmord. 2025. JSesh: An Open-Source Hieroglyphic Text Editor. <https://github.com/rosmord/jsesh>. Accessed: March 27, 2025.
- [9] Maxim Tkachenko, Mikhail Malyuk, Andrey Holmanyuk, and Nikolai Liubimov. 2020–2025. Label Studio: Data labeling software. <https://github.com/HumanSignal/label-studio>. Open source software available from <https://github.com/HumanSignal/label-studio>.